

IB MATH AA SL

Lesson 22

Modelling with Sinusoidal Functions

Trigonometry · 60-minute lesson

Syllabus SL 3.7 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. What is the maximum of $y = \sin x$?
2. What is the minimum of $y = \sin x$?
3. After how many degrees does $\sin x$ repeat?

Answers (click to reveal)

1. 1
2. -1
3. 360°

What we're learning today

LEARNING OBJECTIVES

- 1 Identify**
amplitude, period and vertical shift.
- 2 Interpret**
 $y = a \sin(b(x - c)) + d$.
- 3 Find**
the parameters from a graph or context.
- 4 Use**
a model to make predictions.

SUCCESS CRITERIA — I CAN...

- ✓
find the amplitude a
- ✓
find the period from b
- ✓
find the vertical shift d (the midline)
- ✓
build and use a sinusoidal model

The sinusoidal model

THE PARAMETERS

$$y = a \sin (b(x - c)) + d$$

a

amplitude = (max - min) / 2

d

midline = (max + min) / 2

b

controls the period

PERIOD

Period

= $360^\circ/b$ (or $2\pi/b$)

c

horizontal (phase) shift

max

$d + a$; min: $d - a$

Amplitude & midline

QUESTION

A model $y = a \sin(\dots) + d$ has maximum 17 and minimum 3.

Find a and d .

SOLUTION

Amplitude $a = (\max - \min)/2$:

$$a = \frac{17 - 3}{2} = 7$$

Midline $d = (\max + \min)/2$:

$$d = \frac{17 + 3}{2} = 10$$

Your turn — parameters

ANSWER IN YOUR BOOK

1. Max 9, min 1. Find a and d .
2. A graph repeats every 180° . Find b .
3. Max 5, min -5 . Find the amplitude.

Answers (click to reveal)

1. $a=4$, $d=5$
2. $b = 360/180 = 2$
3. 5

Finding the period

FROM b

$$period = \frac{360^\circ}{b} = \frac{2\pi}{b}$$

Bigger b

shorter period (squashed wave)

From a graph

read one full cycle to get the period, then find b

REAL CONTEXTS

tides, temperature, daylight hours

Ferris wheels, sound waves

d

the average/centre value.

Find b from the period

QUESTION

A sinusoidal model completes one cycle every 12 hours.

Find the value of b (in degrees).

SOLUTION

Period = $360^\circ/b$, and the period is 12:

$$12 = \frac{360^\circ}{b}$$

Rearrange for b :

$$b = \frac{360^\circ}{12} = 30$$

Exam-style question

The depth of water is modelled by $y = a \sin(b(t - c)) + d$.

The maximum depth is 8 m, the minimum is 2 m, and the period is 12 hours.

- (a) Find a . [2]
- (b) Find d . [1]
- (c) Find b (in degrees). [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $a = (8-2)/2 = 3$ (M1)(A1)
- (b) $d = (8+2)/2 = 5$ (A1)
- (c) $b = 360/12 = 30$ (M1)(A1)

Common mistakes



Amplitude

It is HALF the max-min, not the whole range.



Midline

d is the average of max and min, not 0 (unless symmetric).



Period & b

period = $360/b$ — they are inversely related.



Units

Match degrees or radians throughout.

Mixed practice

QUESTIONS

1. Max 12, min 4. Find a and d .
2. Period 90° . Find b .
3. $a = 6$, $d = 10$. Find the max and min.
4. Period $2\pi/3$ (radians). Find b .

Answers (click to reveal)

1. $a=4$, $d=8$
2. $b=4$
3. max 16, min 4
4. $b=3$

Plenary & exit ticket

KEY FORMULAS

$$y = a \sin (b(x - c)) + d$$

$$\text{period} = \frac{360^\circ}{b}$$

EXIT TICKET — before you leave

1. Max 10, min 2. Find the amplitude.
2. Find d if max 10, min 2.
3. Period 60°. Find b.

Answers (click to reveal)

1. 4
2. 6
3. 6

Self-assess:
 • need more help
 • nearly there
 • confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Modelling with sinusoidal functions.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 23

End of Topic Test — Trigonometry. Revise Lessons 17–22.